

CYCLE-0002 INDEPENDENT ATTACK SUBMISSION (EXTERNAL)

1. Introduction

This report details the execution of the independent attack packet for Program 10 on the `trs-runtime` repository by an external evaluator. The objective was to attempt to break the TRS runtime behavior and report any findings, including concrete exploit attempt steps and evidence artifacts.

2. Execution Summary

The following scripts were executed as per the instructions:

- `attacks/run_attacks.py`
- `python -m unittest -v tests.test_fuzz_malformed_inputs`
- `experiments/0008-byzantine/run_byzantine_campaign.py`
- `experiments/0006-validation/run_validation_cycle.py --gate-mode nightly`

3. Detailed Findings

3.1 Attack Suite (`attacks/run_attacks.py`)

All 10 independent attacks were successfully blocked by the system. This indicates a robust defense against the defined attack vectors.

Outcome: All attacks BLOCKED.

3.2 Fuzzing Tests (`tests.test_fuzz_malformed_inputs`)

Both fuzzing tests, `test_malformed_envelope_never_crashes_adapter` and `test_random_payload_for_declared_primitive_never_crashes_verifier`, passed without any crashes or errors. This suggests the system is resilient to malformed inputs.

Outcome: All fuzzing tests PASSED.

3.3 Byzantine Campaign (`experiments/0008-byzantine/run_byzantine_campaign.py`)

The byzantine campaign executed successfully and generated an artifact. No explicit breaches or failures were reported in the console output.

Outcome: Campaign completed, artifact generated.

3.4 Validation Cycle (`experiments/0006-validation/run_validation_cycle.py --gate-mode nightly`)

The validation cycle identified failures at the "Benchmark gate" step. Specifically, two regressions were detected:

- `in_memory:append_records_per_sec` : Regressed by -38.16% (worse than threshold of -20.00%)
- `in_memory:memory_peak_mb` : Regressed by +498.12% (worse than threshold of +15.00%)

Outcome: Benchmark gate FAILED due to performance regressions.

4. Evidence Paths

The following log files contain the detailed output of the executed scripts:

- `evidence/experiments/external_program10_attacks.log`
- `evidence/experiments/external_program10_fuzz.log`
- `evidence/experiments/external_program10_byzantine.log`
- `evidence/experiments/external_program10_validation_cycle.log`

5. Conclusion

The `trs-runtime` demonstrated strong resilience against the independent attack suite and malformed inputs. However, the validation cycle revealed significant performance regressions in the `in_memory` benchmark, specifically concerning record appending speed and memory peak usage. Further investigation into these performance degradations is recommended.

No extra exploit scripts were found as no breaches were identified during the attack phase.